

Principles of Antibiotic Use

Antibiotics are not effective against viral infections such as the common cold or influenza, and prescribing them for viral illness drives antimicrobial resistance. Amoxicillin is the first-line antibiotic for acute otitis media in children when antibiotic treatment is indicated. For patients with a true penicillin allergy, a macrolide such as azithromycin may be used as an alternative.

Adverse Effects and Resistant Organisms

Clostridioides difficile infection is a serious complication of antibiotic therapy, and clindamycin carries one of the highest risks. Oral vancomycin or fidaxomicin are used to treat *C. difficile* infection. Methicillin-resistant *Staphylococcus aureus* (MRSA) infections requiring intravenous therapy are commonly treated with intravenous vancomycin. Fluoroquinolones such as ciprofloxacin carry warnings for tendon rupture, particularly in older adults.